



SE[SUBALG]IndentEndIndent *Indent *EndIndent

TEXT CLASSIFICATION IN NLP: MEDIA BIAS DETECTION, [STUDENT 2 TOPIC], AND [STUDENT 3 TOPIC]

by

ATAKAN GÜL, 121200152
[STUDENT 2 NAME], [STUDENT 2 ID]
[STUDENT 3 NAME], [STUDENT 3 ID]

Submitted for the course

CMPE346: NATURAL LANGUAGE PROCESSING

for the assignment

FINAL PROJECT

June 2026

Abstract

In this paper, we performed text classification on three different NLP tasks using an identical deep learning model. Each task is an application of classification in the NLP field: Media bias detection of news articles, [Student 2 task], and [Student 3 task]. Three models are applied to each task, including one traditional deep learning model Bidirectional LSTM, and two transformer-based pre-trained models BERT and RoBERTa. The three models are then fine-tuned on the dataset for each specific task, and accuracy and weighted F1-score are used as evaluation metrics. For all three tasks, we confirmed that transformer-based models outperform Bidirectional LSTM, and RoBERTa attains the best results.

TABLE OF CONTENTS

ABSTRACT	ii
TABLE OF CONTENTS	iii
INTRODUCTION	1
1 Atakan Gül — Media Bias Detection in News	2
1.1 Methodology	2
1.2 Experiments & Findings	4
1.3 Discussion	5
1.4 Conclusion	6
2 [Student 2 Name] — [Student 2 Topic]	7
2.1 Methodology	7
2.2 Experiments & Findings	8
2.3 Discussion	9
2.4 Conclusion	10
3 [Student 3 Name] — [Student 3 Topic]	11
3.1 Methodology	11
3.2 Experiments & Findings	12
3.3 Discussion	13
3.4 Conclusion	14
References	15

Introduction

Text classification is one of the foundational tasks in NLP and underpins many practical applications we use on a daily basis. At its most basic, text classification assigns a label from a predetermined set to an instance of text based on its contents, providing a pathway to automating understanding over large scales. These applications include, but are not limited to: finding misinformation and media bias in news; classifying the argumentative intent of scientific documents; and classifying harmful content in social media.

While the formulation of text classification may be seemingly simple (the assignment of a discrete label to an item of text), the complexities of human language provide a variety of challenges to be addressed in text classification such as learning the semantics and contextuality of words. Early techniques relied on manually engineered features, shallow machine learning models such as Naive Bayes and Support Vector Machines [5], and more recently recurrent neural networks, particularly Long Short-Term Memory networks [4], began learning representations directly from the text. Most impactfully however, was the introduction of the Transformer [1] and subsequently, the pre-training and subsequent release of very large scale language models such as BERT [2] and RoBERTa [3].

This project looks at text classification in three different NLP tasks each corresponding to a different area of application. The group then proceeds to individually train three different models (BiLSTM, BERT, RoBERTa) on three individually chosen datasets and reports their performance in terms of accuracy and weighted F1-score. The intention of this is to compare how the three models behave under different circumstances.

Chapter 1

Atakan Gül — Media Bias Detection in News

1.1 Methodology

This work targets the task of media bias detection from news articles. It is a classification task where the model has to decide whether a news sentence has biased framing or no bias. Media bias detection is a non-trivial NLP problem because bias is manifested by subtler things like phrasing, framing and tone, and not by stating false facts [6].

In this work, 3 models are tested:

Bidirectional LSTM. A Long Short-Term Memory network is a type of recurrent neural network designed to remember information for long periods of time [4]. Bidirectional LSTMs are trained in both directions in order to make better use of the context. In our work the model takes an embedding layer, two layers of 256-hidden-dimension Bidirectional LSTMs and a linear classification layer on top of the concatenation of the last forward and backward hidden states.

BERT. BERT is a pre-trained language model based on the Transformer encoder architecture. It is trained on a large corpus using a masked language modeling objective and a next sentence prediction objective and therefore is able to produce deep bidirectional context representations [2]. We feed the [CLS] token output through a linear classifier. For the implementation of BERT we use the `bert-base-uncased` variant.

RoBERTa. The Robustly Optimized BERT Pretraining Approach (RoBERTa) is a variant of BERT in which the next sentence prediction task was removed, trained on a larger corpus using larger batch size and uses dynamic masking [3]. This architecture has been shown to perform consistently better on

downstream tasks. For RoBERTa we use the **roberta-base** variant.

The transformer models (BERT and RoBERTa) are fine-tuned from the beginning with an AdamW optimizer and a linear learning rate scheduler with warmup. The LSTM model is trained from scratch using the Adam optimizer.

1.2 Experiments & Findings

Dataset. The BABE dataset (Bias Annotations By Experts) [7] is employed for this task. It is composed of 4,121 news sentences gathered from a diverse collection of sources, each rated as either biased (1) or not biased (0) by trained annotators. It is divided into 3,121 training and 1,000 test samples. Since the labeling is conducted with the help of expert annotators and has been designed carefully, it serves as a reliable reference for media bias detection.

Training Configuration. For all models, 3 epochs of training are performed with early stopping on F1 score on the validation set, a stopping threshold of 0.5% difference between consecutive performances and a patience of 2 epochs. A batch size of 16 and a maximum sequence length of 128 are set. The learning rate for transformers is 2×10^{-5} and for the LSTM is 1×10^{-3} . All models are trained on an NVIDIA GeForce RTX 3060 Laptop GPU.

Results. Table 1.1 presents the performance of each model on the BABE test set.

Model	Accuracy	F1
LSTM	0.6840	0.6772
BERT	0.8120	0.8122
RoBERTa	0.8580	0.8585

Table 1.1: Model performance on the BABE dataset for media bias detection.

1.3 Discussion

From these figures, it is clear that there is a performance hierarchy across the three architectures: The Bidirectional LSTM gets 68.4

BERT scores well, gaining 81.2

RoBERTa scores an excellent 85.8

1.4 Conclusion

This paper applied three deep learning architectures to detect media bias using the BABE dataset. Experiments indicated that the pretrained transformers substantially outperformed the LSTM baseline, and RoBERTa produced the best results with accuracy 85.8% and weighted F1 0.858. The experimental results also indicate the crucial role of pretraining for low-resource classification problems since it is not possible to learn the nuances of language for news framing and bias without pretraining. Further investigations could use larger transformer models or domain adaptation pretraining on news corpora.

Chapter 2

[Student 2 Name] — [Student 2
Topic]

2.1 Methodology

[Student 2 fills this in]

2.2 Experiments & Findings

[Student 2 fills this in]

2.3 Discussion

[Student 2 fills this in]

2.4 Conclusion

[Student 2 fills this in]

Chapter 3

[Student 3 Name] — [Student 3
Topic]

3.1 Methodology

[Student 3 fills this in]

3.2 Experiments & Findings

[Student 3 fills this in]

3.3 Discussion

[Student 3 fills this in]

3.4 Conclusion

[Student 3 fills this in]

Bibliography

- [1] A. Vaswani et al., “Attention is all you need,” *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [2] J. Devlin, M. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL*, 2019, pp. 4171–4186.
- [3] Y. Liu et al., “RoBERTa: A robustly optimized BERT pretraining approach,” *arXiv preprint arXiv:1907.11692*, 2019.
- [4] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [5] T. Joachims, “Text categorization with support vector machines: Learning with many relevant features,” in *Proc. ECML*, 1998, pp. 137–142.
- [6] F. Hamborg, K. Donnay, and B. Gipp, “Automated identification of media bias in news articles: An interdisciplinary literature review,” *International Journal on Digital Libraries*, vol. 20, pp. 391–415, 2019.
- [7] T. Spinde et al., “Neural media bias detection using distant supervision with BABE,” in *Findings of EMNLP*, 2021.